The cells of many of the organisms are similar in size. Almost all cells maintain uniformity in size (1-2 μm in diameter) and the bigger ones may be just 5-10 times larger. Why is such uniformity in cell size maintained? The surface/volume ratio for any object of any shape depends on its size. The complex biochemistry that takes place inside the cell requires a significant volume and exchange with the external environment. If the size is too large, then there will not be enough surface for the exchange of substances with its environment and the process will not occur. Thus, the size relates to the biochemical process i.e metabolism. An unicellular bacterial cell has smaller size than that of the cells of higher organisms as their metabolism is simple while a virion(viral particle) is smaller than bacteria as they depend on their host for their survival. Yet, there are unusual sizes such as very small bacteria which are 0.2 μm in length, while cells of nervous systems of vertebrates are about a metre long.

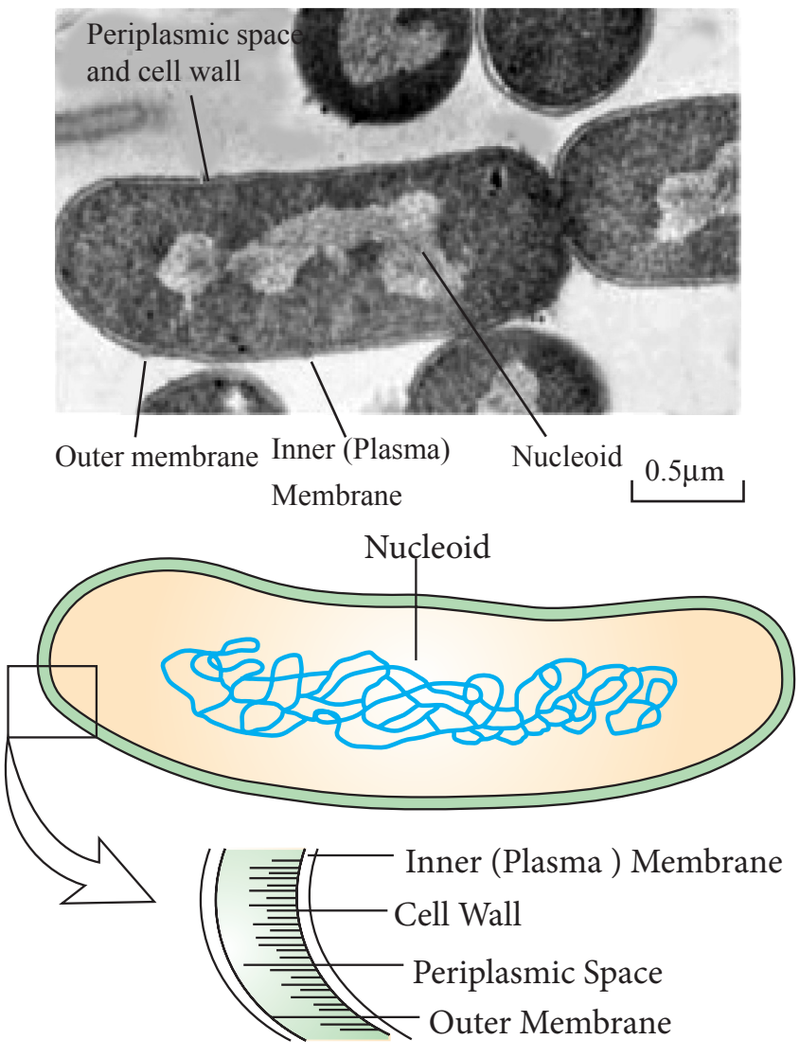


Figure 1.1 Structural organization of the prokaryotic cell

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